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Benchmark Performance Without External Validity: A Systematic Review of AI-Based Fake Job Detection (2017–2026)

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ABSTRACT: Fraudulent job posting detection models can be found since 2017 and many report an accuracy of more than 95%. This systematic review addresses the question: Do those results apply to other datasets? Studies were identified between January 2017 and March 2026 by conducting searches in five databases between January and March 2026. After duplicate records were removed, there were 89 records and 28 met inclusion criteria, with 54 being read in full. An addendum search was conducted for additional studies in early 2026 and resulted in 2 more studies added for a total of 30 studies, with 26 reporting extractable performance data. A seven dimension risk-of-bias rubric was used to rate quality. The data for 54% of the studies with extractable results are provided by One benchmark, the Employment Scam Aegean Dataset, which was collected between 2012 and 2014. In this dataset, a classifier which simply classifies every posting as legitimate achieves a 95.2% score, many reported accuracies are within a few points of this, and only half of the studies even report fraud-class metrics like precision, recall or F1. There is one report of external validation and no report of temporal or production validation. Most of the evidence of large language models altering the threat model is seen in adjacent tasks like artificial intelligence text detection, and not in job-fraud data. They were screened by one reviewer, records were only searched from English, and not kept per database. Accuracy figures of the headlines are to be interpreted as maximum values. The most pressing need is a valid and contemporary multilingual benchmark with temporal divisions and assessment measures which prevent the occurrence of data leakage.

INDEX TERMS: Benchmark validity, class imbalance, evaluation methodology, fake job detection, large language models, online recruitment fraud, systematic review.

I. INTRODUCTION

A. Background

Web-based hiring platforms account for a big portion of job search and recruitment, and the infrastructure that is actually cheaper for job seekers to locate job openings is the same as it is cheaper for fraudsters to reach them. There are several types of recruitment fraud, such as asking for up front fees, taking identity documents, and in the worst recorded instances, trafficking disguised as a job overseas. In the U.S., there were approximately 90 million reported losses in 2020 but that rose to about $501 million in 2024, and the number of reported losses tripled over that period [1]. In 2025, the total amount of money lost due to fraud was reported to have reached $15.9 billion [2] and in separate reports, the Federal Trade Commission has outlined a growing trend in "game-like job" scams [3] and [4]. These figures are only background, and no analysis in this review is based upon the use of these figures.

So, there is a cost to both sides when trying to detect fraudulent postings, making this an applied machine learning problem. A platform must filter high numbers of postings, and the loss of fraud as well as the loss of legitimate employers is a bad thing. In 2017 Vidros et al. presented the Employment Scam Aegean Dataset (EMSCAD) [5]. Since then, the literature has progressed from classical classifiers to fine-tuned transformer encoders and, more recently, systems which have adopted large language models (LLMs) or systems which must defend against LLMs.

B. Rationale for This Review

There are some method families and recurring issues in this field that have been summarized in a recent review [6] such as class imbalance, limited benchmarking, etc. There are three observations that inspire a new synthesis. Firstly, studies using transformers since 2023 claim accuracy of up to 99% [7] for a benchmark where always predicting the majority class yields accuracy of 95.2%, and therefore a closer examination of the results in those studies is warranted. Second, it is reduced the price of writing convincing posting text, and the strength of the text detectors against LLM rewriting is measured in neighboring domains. Third, there are no previous studies addressing evaluation validity as its primary focus of study, especially in the context of the possibility of data leakage. It is no longer the question of whether some classifier achieves high accuracy on EMSCAD, but rather why that is. It's about whether the performance that was reported can be used by a platform when it is putting things into action.

C. Contributions

There are four contributions that the review makes. (1) A taxonomy of the 30 studies included, in which the inputs and outputs of the studies are represented as a five-dimensional grid, and empty cells indicate that a specific cell in the design space has not been studied (2) An evaluation study about the validity of evaluations, including the handling of class imbalance, metric completeness, data leakage risk and benchmark dependence, showing that external validation is not found in the literature. (3) Adopting separation between direct evidence of job-fraud and transferable evidence of its adjacent domain to detect text generated by LLMs, so that robustness claims are not made for the job-fraud domain without evidence. (4) A research agenda with each gap identified and

accompanied by evidence of support, the implication of the gap, methodology required and a design of the study.

D. Scope

The review includes empirical research on automated detection of fraudulent job postings, from January 2017 to March 2026, in English. It does not include purely rule-based systems (no learning component), general fraud detection (without job-posting focus) and opinion pieces without empirical evaluation. The limitation of the English-only rule is that recruitment fraud is a multilingual issue.

II. Review Questions

RQ1: What datasets have been used for detecting fake job postings and what input modalities and fraud representations have been used?

RQ2: Which machine learning, deep learning, transformer and LLM-based approaches are evaluated, and where are they applied in learning?

RQ3: How are the reported results similar and methodologically reliable, when considering validation protocols, handling of class imbalance and leakage risk?

RQ4: Does there exist evidence for temporal, cross-platform, multilingual and adversarial generalization exist?

RQ5: What are the gaps that the evidence addresses, and what would be the study designs to address these gaps?

III. Methodology

A. Protocol

The review was based on a protocol established prior to the screening process, which included research questions, inclusion and exclusion criteria, search strategy, risk-of-bias criteria and extraction fields. Reporting is based on the PRISMA 2020 statement and explanation document [8, 9] with some adaptations that make a strict compliance not possible: screening was performed by one reviewer, without further checks from a second reviewer, and the number of records retrieved per database from the search log was not saved. These are both mentioned in Section X and Appendix B provides a search string and re-screening protocol.

B. Information Sources and Search Strategy

The library will supply information on how to search for information as well as information sources. The searching of the sources was conducted from January to March 2026 and it involved IEEE Xplore, ACM Digital Library, Google Scholar, ScienceDirect, and arXiv. The query was a combination of two concept blocks: the fraud target and the method family. Two concept blocks were combined: the concept block of frauds targeted, which included online recruitment fraud, employment scam, fake job posting, fraudulent job advertisement, and the concept block of method family, which included machine learning, deep learning, neural network, natural language processing, classification, detection, BERT, transformer, LSTM, large language model. The totals for each screening stage are provided in table 1 of appendix B. The table does not contain any database-specific information as the counts for each stage are not saved.

C. Eligibility Criteria

The study was included if it (i) was published between January 2017 and March 2026, (ii) reported an empirical evaluation, (iii) focused on detection of job fraud, (iv) employed a learning-based approach (either supervised or semi-supervised, unsupervised or LLM-based), and (v) provided sufficient methodological information to evaluate contributions. Studies were excluded if they were opinion or position papers, duplicated or were greatly overlapping with other reports, had less than 100 examples, did not include an evaluation of the results, or were not conducted in English.

D. Study Selection

142 records were found. With 53 duplicate records removed, 89 records were screened on title/abstract and 35 excluded. Fifty-four reports were read in full text, while 26 reports were excluded: 8 were not specific to job postings, 7 lacked an empirical evaluation, 6 provided inadequate methodological details and 5 duplicated more than 80% of another report included. This left 28 studies. In March 2026, two more studies were identified through citation chasing after the initial searches had been conducted, and in addition, one more study was identified and added via a documented addendum, to reach 30 (142 − 53 − 35 − 26 = 28; 28 + 2 = 30). Fig. 1 shows the flow.

TABLE 1. Search and screening statistics.

Stage	Records	Note
Records identified (5 databases, Jan–Mar 2026)	142	Per-database counts not retained
After duplicate removal	89	53 duplicates removed
Excluded at title/abstract screening	35	Reasons recorded in screening log
Assessed at full text	54	
Excluded at full text	26	8 not job-specific; 7 no empirical evaluation; 6 insufficient detail; 5 overlapping
Included (main search)	28	
Addendum studies (Q1 2026, citation chasing)	2	Added March 2026
Total included	30	26 with extractable performance data

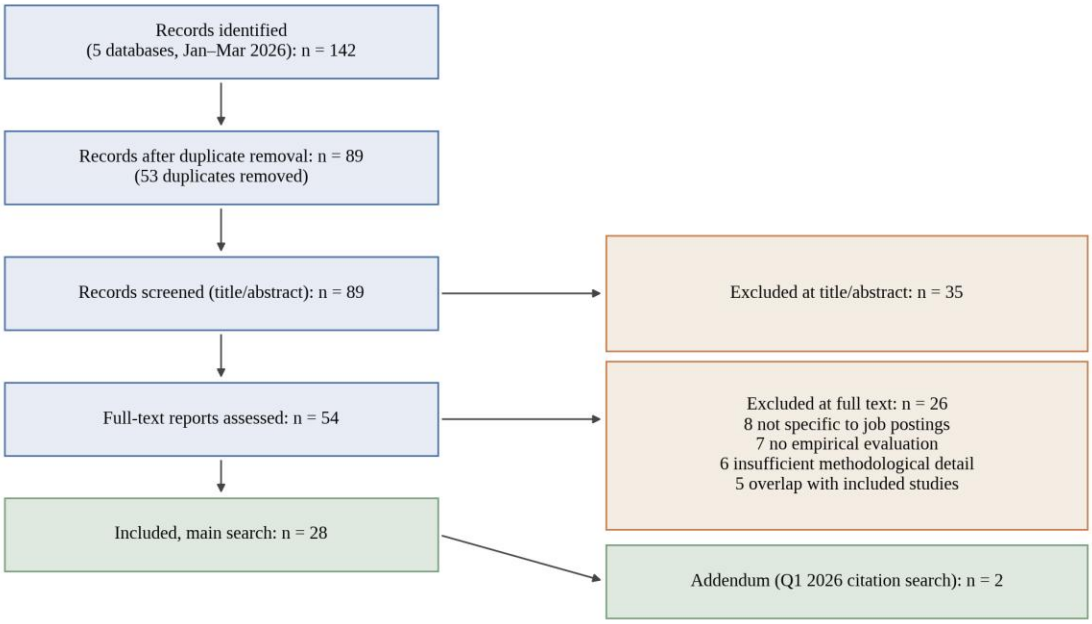


FIGURE 1. Study selection flow. The two addendum studies from the March 2026 citation search are shown separately from the main flow.

E. Risk-of-Bias and Quality Assessment

A domain-specific risk-of-bias rubric was used to evaluate the quality of the study, instead of a generic quality score, because a generic score will "reward" surprising findings rather than sound ones. The rubric consists of seven dimensions: documentations of the size, provenance, and class-distribution of the datasets (low risk); handling the class imbalance dimension (low risk); no temporal or external validation rates describes the validation protocol (high risk); the completeness of metrics (accuracy alone rates as low risk); fitting the validation protocol before the split dimension (preprocessing or resampling before splitting, or no description of the train/test protocol, rates as high risk); documentation of the availability and reporting of hyperparameters, as well as the availability of code or data, rates as a dimension of reproducibility (low risk); and the description of a simple baseline or comparison (at least one simple baseline as unclear, or no description of the train/test protocol, rates as high risk). It is recommended to use decision rules as outlined in Appendix C. All ratings were made by author. Since two independent raters were not available, no inter-rater statistic is reported.

F. Data Extraction and Synthesis

The following was extracted from each study: bibliographic information, dataset, size of the dataset, fraud-class share, language, features, model, pre-processing, sampling strategy, validation protocol and all reported metrics (see Appendix A). When values are not reported in a study they are not estimated and are recorded as NR (not reported). You don't need to be doing a meta-analysis; synthesis is narrative and structured. Pooled means and accuracy confidence intervals are not meaningful due to the variation in the studies with respect to datasets, splits, data preprocessing, and class distributions. The performance is therefore reported in terms of the study and compared with each other only on the common basis of evaluation conditions (Section V).

IV. Results

A. Study Characteristics

There are 30 studies (28 of the main search and 2 from the addendum), from which 26 report extractable performance data. Fig. 2 depicts the publication year of modeling family. Every study up until 2020 is represented by classical classifiers, which are still the biggest group. Deep learning models start to emerge in 2021 and transformer based work and LLM in 2025. A taxonomy of the five dimensions used to classify the corpus is provided in Table 2 and the number and type of datasets, the model families, the validation protocols, key reported results, and the overall risk-of-bias rating for all the included studies are summarized in Table 3.

TABLE 2. Five-dimension taxonomy used to organize the included studies.

Dimension	Categories	Coverage in the corpus
Input modality	Text; metadata; structure; temporal; behavioral	Text with metadata is most common; temporal and behavioral signals are almost unused
Modeling paradigm	Classical ML; deep learning; transformer encoder; LLM-based	Classical methods are most frequent; transformer and LLM studies appear from 2025
Learning setting	Supervised; semi-supervised; unsupervised; zero-/few-shot	Supervised classification dominates; label-free formulations are speculative
Threat model	Human-written; LLM-generated; paraphrased; human–AI hybrid	Almost entirely human-written fraud
Validation setting	Random split; cross-validation; external dataset; temporal; production	External validation in 1 study; temporal or production in none

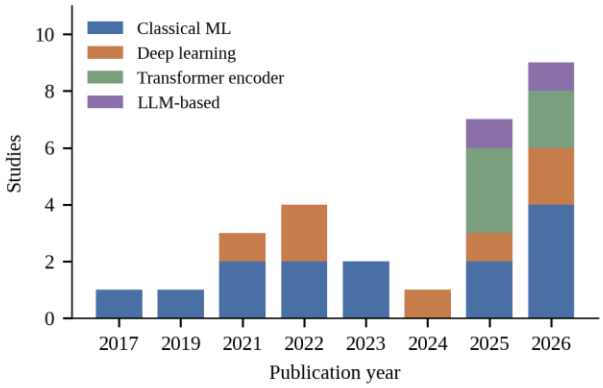


FIGURE 2. Included studies by publication year and modeling family.

TABLE 3. Characteristics and risk ratings of included studies. NR = not reported; V = validation protocol (CV = cross-validation, Split = single train/test split, Bench. = benchmark evaluation); Acc = accuracy as reported by the study.

ID	Study	Year	Data	Family	V	Key reported result	Risk
S1	Vidros et al. [5]	2017	EMSCAD	Classical	CV	Dataset characterization; baselines	Low
S2	Alghamdi, Alharby [10]	2019	EMSCAD	RF ensemble	NR	Acc 97.41%; profile, logo, industry top features	Medium
S3	Habiba et al. [11]	2021	EMSCAD	Classical + DL	NR	Comparative classifier study	Medium
S4	Lokku et al. [12]	2021	EMSCAD	RF + SMOTE	NR	TF-IDF, SMOTE, RF	Medium
S5	Divya, Banik [13]	2021	NR	BiLSTM	NR	BiLSTM with word embeddings	High
S6	Khandagale et al. [14]	2022	EMSCAD	Ensemble	NR	Ensemble classification	Medium
S7	Naudé et al. [15]	2022	Public corpus	GB + features	CV	Fraud-type classification, F1 0.88	Low
S8	Alandjani [16]	2022	NR	Classical	NR	Multi-classifier comparison	High
S9	Wang et al. [17]	2022	NR	Classical + DL	Split	TextCNN best, metrics above 0.93 (balanced)	Low
S10	Udayshankar et al. [18]	2023	EMSCAD	DNN	NR	Acc about 98%	Medium
S11	Reddy et al. [19]	2023	EMSCAD	Ensembles	Split	XGBoost 96.8%; five ensembles compared	Low
S12	Vu et al. [20]	2024	NR	DL NLP	NR	Word2Vec + metadata framework	Medium
S13	Medapati et al. [21]	2025	EMSCAD	Ensemble	NR	Boosting comparison	Medium
S14	Itnal [22]	2025	Kaggle mirror	RF	Split	RF best among four classifiers	Medium
S15	Pradeep [23]	2025	Custom (5,000)	XGBoost	NR	Acc 94.2%; SHAP explanations	Medium
S16	Krishna et al. [24]	2025	EMSCAD	XGBoost	NR	Acc about 98%	Medium
S17	Reddy et al. [25]	2025	EMSCAD	Boosting	NR	XGBoost, RF, LR comparison	Medium
S18	Taneja et al. [7]	2025	EMSCAD	BERT	Split	Acc 99%; F1 0.93	Low
S19	Yang et al. [26]	2025	Fraud-R1 (8,564)	LLM evaluation	Bench.	Fake-job inducements hardest; EN–ZH gap	Low
S20	Wu et al. [27]	2025	Simulation	LLM agents	Bench.	Deceptive intent common (adjacent domain)	Low
S21	Harika et al. [28]	2026	EMSCAD	ANN/CNN/RF	NR	Acc about 90%	High
S22	Tripathi [29]	2026	EMSCAD	LightGBM/BiLSTM	Split	Acc 98.18% / 98.71%; AUC 0.91	Medium
S23	Nivetha et al. [30]	2026	Indian platform	XLM-R	Split	XLM-R best; SHAP/LIME explanations	Medium
S24	Ashika, Victoire [31]	2026	NR	Llama-3 + ML	NR	LLM–ML hybrid pipeline	High
S25	Swamy, Sammulal [32]	2026	NR	LSTM + ensemble	NR	Acc about 98% (self-reported)	High
S26	Lakshmi et al. [33]	2026	NR	SVM	NR	SVM best; Flask deployment	High
S27	Ivaranjani et al. [34]	2026	NR	GPT-2/XLNet/LSTM	NR	Transformer comparison	Medium
S28	Perrone, Romano [35]	2026	ARB (adjacent)	Detector evaluation	Bench.	60–78 pt recall drop on LLM rewriting	Low
S29	Li et al. [36]	2026	Multimedia (adjacent)	Stylistic profiling	Bench.	Cross-domain robustness gain	Low
S30	Kulatilleke et al. [37]	2026	AIGT (adjacent)	Contrastive	Bench.	Robust to 90% perturbation intensity	Low

Risk ratings follow the rubric in Appendix C. Studies S19, S20, and S28–S30 are adjacent-domain evidence analyzed only in Section VI; they are included in the corpus for the LLM-era synthesis but excluded from recruitment-fraud performance counts.

B. Datasets (RQ1)

The evidence base is almost exclusively comprised of EMSCAD [5] research. A total of 26 studies had extractable data, of which 14 relied on EMSCAD data and 9 relied on custom or platform-specific data, with 3 studies not clearly identifying their data source (Fig. 3). EMSCAD has 17,880 postings from 2012-2014 from one applicant-tracking system, with 866 being fraudulent (4.8:95.2 [5]). Most of the custom datasets are small and not well documented to allow for cross-study comparisons. If the source study does not provide these data, they are reported as NR in Table 4.

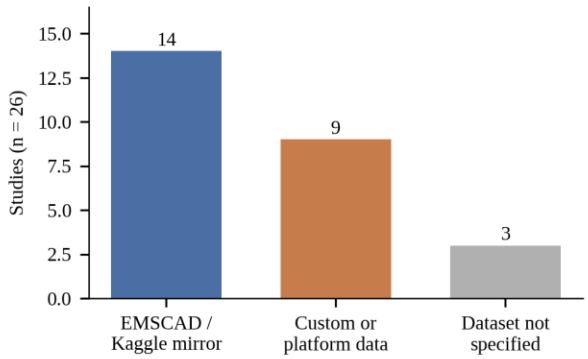


FIGURE 3. Dataset usage across the 26 studies with extractable data.

TABLE 4. Dataset and benchmark properties.

Property	EMSCAD [5]	Custom datasets (typical reporting)
Size	17,880 postings	Mostly under 20,000; several under 6,000; some NR
Fraud class share	866 (4.8%)	Rarely stated; NR in most studies
Collection period	2012–2014	Usually recent but rarely dated
Language	English	Mostly English; one multilingual study [30]
Source platform	One ATS platform (Workable)	Single platform or unspecified
Metadata fields	Rich (logo, profile, salary, etc.)	Varies; often text only
Public availability	Public (Kaggle mirror)	Rarely released
Class imbalance	Yes (about 95:5)	Present where stated

Inputs fall into five groups: text (title, description, requirements, benefits), metadata (company logo, company profile, salary, employment type, telecommuting flag), structure (description length, formatting), temporal signals (account age, posting dates), and behavioral signals. Text combined with metadata is the most common setup. No included study uses network or behavioral signals at scale, and temporal features are almost entirely unused.

C. Modeling Approaches (RQ2)

Table 5 counts the 26 extractable studies by model family. Classical methods remain the most frequent family. Random Forest [38] is the most common single classifier, usually paired with TF-IDF [39] text features and metadata. Alghamdi and Alharby report 97.41% accuracy for a random-forest-based ensemble on EMSCAD, with company profile, logo, and industry as the leading features [10]. Boosting and ensemble methods are also common. A comparison of Random Forest, LightGBM [40], XGBoost [41], CatBoost, and Extra Trees found 96.8% accuracy for XGBoost on EMSCAD [19], and several 2025–2026 studies report 94% to 98% for boosting and ensemble pipelines [21], [22], [24], [29], [32].

Deep learning studies use DNNs (about 98% accuracy on EMSCAD [18]), CNNs, and recurrent models. A controlled comparison of four classical and three deep models on balanced and imbalanced variants of the data found TextCNN strongest, with all metrics above 0.93 on the balanced set [17]. BiLSTM [42] models with word embeddings [43]–[45] are the most common recurrent choice [13], [20].

Transformer encoders [46], such as BERT [47] and its distilled variant DistilBERT [48], mark the current performance frontier. Fraud-BERT, a fine-tuned BERT, reports 99% accuracy and an F1 of 0.93 on EMSCAD [7], the strongest class-balanced result in the corpus. A 2026 supervised study reports LightGBM at 98.18% accuracy (ROC-AUC 0.91) and BiLSTM at 98.71% on the same benchmark [29]. Multilingual transformer work has started: a framework for Indian recruitment platforms uses mBERT and XLM-RoBERTa [49] with SHAP [50] and LIME [51] explanations and reports XLM-RoBERTa ahead of monolingual and classical baselines [30].

LLM-based approaches fall into two groups: LLM components inside detection pipelines, such as Llama-3 combined with conventional classifiers [31] and zero- or few-shot GPT screening, and studies that treat LLM robustness as the evaluation target, such as Fraud-R1 [26]. Section VI treats them separately because their evidentiary status differs.

TABLE 5. Models and features across the corpus. Counts refer to the 26 extractable recruitment-fraud studies; a study may appear in more than one row.

Model family	Studies	Typical features
Random Forest	6	TF-IDF + metadata (logo, profile, salary)
XGBoost / LightGBM / CatBoost / other boosting	6	TF-IDF, engineered features, SHAP explanations
DNN / MLP / ANN	3	Embeddings, TF-IDF
CNN / TextCNN	2	Embeddings
LSTM / BiLSTM (+ attention)	4	Word embeddings, metadata
BERT family (fine-tuned)	2	Contextual embeddings; domain fine-tuning
LLM-based (GPT screening, Llama-3 hybrids, Fraud-R1)	3	Zero-/few-shot prompting, hybrid pipelines
Multilingual transformers (mBERT, XLM-R)	1	Multilingual contextual embeddings

D. Evaluation Practices (RQ3)

Fig. 4 and Table 6 summarize which practices the 26 studies report. Thirteen report class-balanced metrics such as F1, precision, or recall alongside accuracy. Nine state the class distribution of their data, six describe the train/test protocol well enough to replicate it, and four report an AUC. One reports external or cross-dataset validation, and none reports temporal or production validation. Some studies report accuracy alone. With a 95:5 imbalance, accuracy without fraud-class recall cannot be interpreted: a classifier that labels every posting legitimate scores 95.2% on EMSCAD and detects no fraud. Precision-recall analysis is generally more informative than ROC analysis on imbalanced data [52], yet only one study reports it.

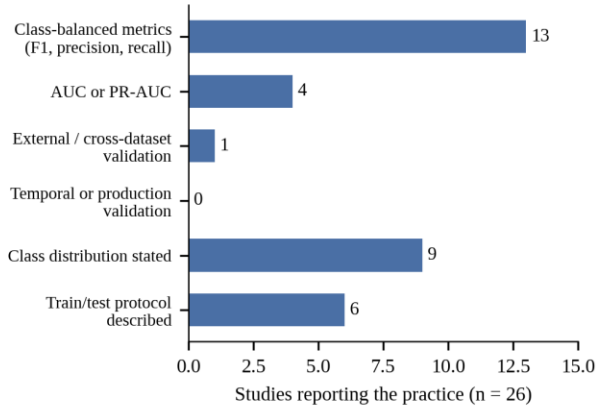


FIGURE 4. Evaluation practices reported across the 26 studies with extractable data. External validation is reported by one study, and temporal or production validation by none.

TABLE 6. Evaluation methodology and metrics as reported by the 26 extractable studies.

Practice	Studies	Notes
Accuracy	24 (92%)	Nearly universal; not interpretable alone at 95:5 imbalance
F1 / precision / recall	13 (50%)	Minimum adequate class-balanced reporting
ROC-AUC / PR-AUC	4 (15%)	One study reports a precision-recall analysis
Class distribution stated	9 (35%)	Others leave the imbalance unclear
Train/test protocol described	6 (23%)	Prerequisite for assessing leakage risk
SMOTE / class weighting / focal loss	9 (35%)	Handling reported unevenly
Cross-dataset or external validation	1 (4%)	Reported by a single study
Temporal / production validation	0 (0%)	Absent from the corpus

V. Critical Evidence Synthesis

A. Comparability of Reported Performance

The range of accuracy reported from the reviewed studies is approximately 84% to 99%, but the accuracy cannot be compared between studies. They are obtained from various datasets, splits, preprocessing operations, class distributions and metrics. The highest values are grouped around EMSCAD, with the percentages for the majority class baseline being 95.2%. Some results from deep learning models are at or below that baseline, meaning that they cannot detect at all other than always predicting "legitimate." For these reasons, no ranking by the accuracy of the models in each cross-study is presented here, nor is it recommended for future work. Only within-study comparisons based on a common split (as in [17] for the classical vs deep comparison and in [19] for the ensemble comparison) can serve as a valid foundation for conclusions about method families.

B. Benchmark Dependence and External Validity

26 studies have been extracted of which 14 are on EMSCAD or its mirror. The data set was gathered in 2012-2014, and fraud has evolved since then: As reported by regulators, frauds now include game-like tasks [3], [4] and posting text is now relatively easy to fake with LLMs. While the accuracy measure applies to the decade-old, single-platform, single-language collection, it does not describe a platform that is today facing fraud. Benchmark saturation would have the accuracy scores steadily increasing on the same benchmark and would not signal an improvement in field detection. A modern benchmark with temporal splits, multiple platforms and LLM-generated fraud is required and new studies should routinely report results across multiple datasets.

Fig. 5 illustrates the changes in data use over time. EMSCAD accounts for 2 of 2 studies in 2017–2020, 5 of 9 in 2021–2023, and 7 of 15 in 2024–2026, so its share is falling as recent studies turn to custom data. The transition also has its price tag: when there are no common standards, it will no longer be possible to compare the results of various studies.

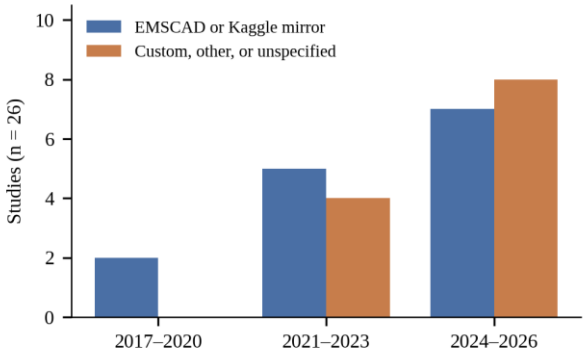


FIGURE 5. EMSCAD or mirror usage versus custom, other, or unspecified data, by publication period (n = 26 extractable studies).

C. Class Imbalance and Metric Practice

The 95:5 imbalance is the hallmark of this task, which the reviewed studies deal with unevenly. Several use SMOTE [53] or a weight on the classes (neither one nor the other), and report on the impact; a few report no explicit handling; and a few use focal loss [54]. The difference between the imbalanced-aware and naive configurations is usually greater than a few points, which is in line with the general literature on imbalanced classification [55], [56]. The bigger issue is reporting. A study that claims detection capability on this data must include an accuracy claim and not present fraud-class recall, precision or a precision-recall plot. These studies are classified as high risk in metric completeness in Table 7 and the accuracy figures should be taken with this qualification.

D. Evaluation Validity and Data Leakage

There are several ways to increase the reported performance that are not apparent in the published figures. Although they are not given many names in the literature reviewed, their names are left open. (i) Resampling prior to train/test split: When SMOTE or other oversampling techniques are applied prior to train/test split, synthetic neighbours to test examples also become a part of training set. (ii) Preprocessing on the whole corpus: TF-IDF vocabularies or statistics for scaling are computed on the whole corpus and test-set information is passed to the training. Random splitting of temporally correlated data: the data from the same period and campaign appears on both sides of the split, which allows that campaign-

specific artifacts can be memorized. (iv) Split-to-split near duplicates that are likely to happen if the corpus consists entirely from a single platform and the same advertisement is posted in different forms on that platform. (v) Tuning on test performance, which can never be ruled out in studies that report one test number and no set of validation numbers.

A total of 26 extractable studies were analyzed, and 6 of them provide sufficient information on train/test to evaluate these risks. In Table 7, leakage risk is unclear or high for the others. These are risk assessments based on reporting, not on particular studies that demonstrate leakage, and only a few studies report the checks that would lead to a determination of the leakage. Until the accuracy of the protocol is improved, it is best read as an upper bound of reported accuracy in this domain.

TABLE 7. Risk-of-bias ratings by study (L = low, U = unclear, H = high risk). Columns: dataset adequacy, imbalance handling, validation, metrics, leakage, reproducibility, baseline, and overall.

ID	Data	Imb.	Val.	Met.	Leak.	Rep.	Base.	Overall
S1	L	L	L	L	L	L	L	Low
S2	L	U	U	U	U	U	L	Medium
S3	L	U	U	U	U	U	L	Medium
S4	U	L	U	U	U	U	U	Medium
S5	H	U	U	U	H	H	H	High
S6	U	U	U	U	U	U	U	Medium
S7	L	L	L	L	L	L	L	Low
S8	H	U	H	U	H	H	H	High
S9	L	L	L	L	L	L	L	Low
S10	L	U	U	U	U	U	L	Medium
S11	L	L	L	L	L	L	L	Low
S12	U	U	U	U	U	U	U	Medium
S13	L	U	U	U	U	U	L	Medium
S14	U	U	U	U	U	U	U	Medium
S15	L	U	U	L	U	L	L	Medium
S16	L	U	U	U	U	U	L	Medium
S17	L	U	U	U	U	U	L	Medium
S18	L	L	L	L	L	L	L	Low
S19	L	n/a	L	L	L	L	L	Low
S20	L	n/a	L	L	L	L	L	Low
S21	H	U	U	H	H	H	H	High
S22	L	U	U	L	U	L	L	Medium
S23	L	U	U	L	U	L	L	Medium
S24	H	U	H	U	H	H	H	High
S25	H	U	H	U	H	H	H	High
S26	H	U	H	U	H	H	H	High
S27	U	U	U	U	U	U	U	Medium
S28	L	n/a	L	L	L	L	L	Low
S29	L	n/a	L	L	L	L	L	Low
S30	L	n/a	L	L	L	L	L	Low

Overall rating is the highest-risk rating among the seven dimensions. Definitions are in Appendix C. n/a = not applicable.

E. Generalization Evidence (RQ4)

There is weak evidence of generalization in each of the directions studied. There is no included study that evaluates on data not collected during the training time, nor is there any that evaluates across platforms or regions. There are anecdotal accounts of models trained with EMSCAD who required further training with regional data, but these are not included here as evidence. One study directly focuses on a multilingual setting [30] and another paper, Fraud-R1, shows significant difference in the understanding of fraud between English and Chinese in LLMs [26]. There is no included job-fraud study that compares against adversarial altered postings; the quantitative evidence for paraphrase susceptibility is provided by AI-text-detection benchmarks in other fields (Section VI-A). For the question RQ4, the answer is therefore that there is almost no direct evidence and this is the key finding for this question.

VI. LLM-Era Recruitment Fraud

Evidence in this section is separated by provenance. Direct evidence comes from studies of recruitment fraud, or of fraud corpora that include fake job postings. Adjacent evidence comes from AI-text detection and adversarial robustness research in other domains. Adjacent results suggest hypotheses about job fraud, but none has been demonstrated on a job-fraud benchmark.

A. LLMs as Fraud Generators

Direct evidence is partial. Fraud-R1 evaluates how well LLMs resist fraud inducements across 8,564 cases drawn from phishing, fake job postings, social media, and news, covering five fraud types and a multi-round protocol with credibility-building, urgency-creation, and emotional-manipulation stages [26]. Three of its findings bear on this review: fake job postings are among the hardest categories, role-play agent settings reduce resistance further, and performance differs substantially between English and Chinese.

The adjacent evidence comes from four sources. A multi-agent simulation study of AI deception finds that deceptive intent is common in goal-driven agent interactions [27]. A benchmark of AI-text detectors shows that detectors trained to separate human from LLM text lose most of their recall when human text is rewritten by an LLM, with drops of 60 to 78 percentage points at a 1% false-positive rate [35]. Stylistic-consistency profiling improves cross-domain robustness [36], and contrastive stylistic regularization stays robust under heavy character- and word-level perturbation [37].

Taken together, the adjacent results make it likely that detectors tuned on human-written fraud will degrade against LLM-written or LLM-polished postings. That has not been measured on a job-fraud benchmark, so LLM-era robustness is treated here as an emerging gap with strong indirect evidence, not as an established one. Fig. 6 shows how the evidence is structured.

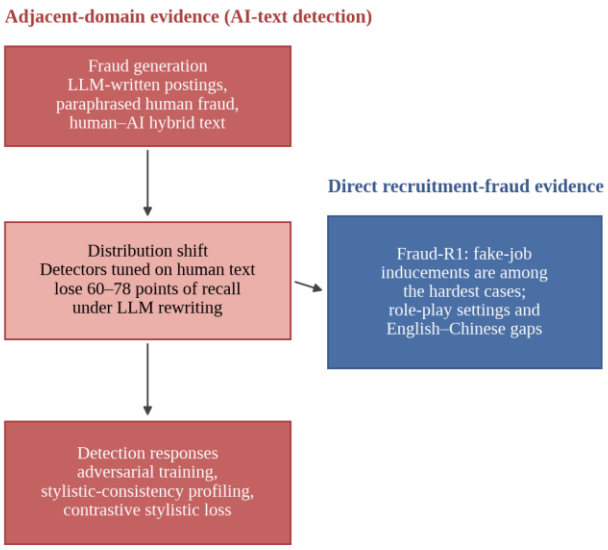


FIGURE 6. Evidence structure for LLM-era fraud and detection. Red elements are adjacent-domain findings; blue elements are direct recruitment-fraud findings.

B. LLMs as Detectors

In the provided corpus, LLM applications in detection are primarily at the pipeline level, including LLM components integrated with classical classifiers [31], transformer models contrasted with the use of LLM-family generators like GPT-2 [34] and zero or few-shot screening when labeled data is limited. A few studies have been conducted with zero/few shot screening, which shows lower accuracy compared to fine-tuned specialist models, but with small numbers and with varying conditions. Currently, there is some evidence that LLMs can be used as parts and label-support tools for human review. It is not an alternative to optimized detectors.

C. Threat Model Implications

The threat-model dimension of the taxonomy (Table 2) separates human-written, LLM-generated, paraphrased, and human–AI hybrid fraud. The corpus concentrates almost entirely on the first category, and the other three are essentially unmeasured for recruitment fraud. This is the clearest case of a gap where the adjacent literature indicates both that the risk is real and that current evaluation practice cannot detect it.

VII. Research Gaps

Table 8 provides a summary of each gap, along with the evidence and consequences, a methodological need, and proposed research, and categorizes the gap as well established, emerging or speculative. The evidence column refers to only what the studies included and surrounding the table have established and where there is a gap in interpretation, the table states so.

Firm foundations are available from the studies included, with the first being dependence on a single aging metric (Section V-B); the second, the lack of external, temporal and production validation (Section V-A and V-E); the third, inconsistent handling and reporting of imbalances (Section V-C); and the fourth, opaque protocols with unquantifiable leakage risk (Section V-D). The emerging gaps are based on adjacent-domain evidence or very recent direct studies: LLM-generated and paraphrased frauds (Section VI), multilingual detection [26] and [30] and explainability beyond explaining feature importance [57] in which only a small number of studies provide any kind of interpretation mechanism. Speculative directions are motivated but not yet supported by evidence: Federated Cross-platform Detection, Proactive Screening Prior to Publication, and One-class Anomaly Formulations requiring no fraud labels.

TABLE 8. Research gaps with evidence basis.

Gap	Evidence of gap	Consequence	Need and proposed research	Status
Benchmark stagnation	14/26 studies on a 2012–2014 collection; accuracy trending toward ceiling	Reported accuracy reflects the benchmark, not field performance	New multi-platform, temporally split, multilingual benchmark with LLM-generated fraud and recorded provenance	Well-established
No external, temporal, or production validation	1 study with any external validation; 0 temporal or production	No evidence that any reported system works outside its own test split	Platform-partnered validation; rolling temporal evaluation	Well-established
Imbalance and metric practice	50% report class-balanced metrics; several report accuracy alone	Uninterpretable or inflated performance claims	Mandatory minority-class metric reporting; PR-AUC	Well-established
Leakage and protocol opacity	6/26 describe the train/test protocol	Leakage cannot be excluded; accuracy is an upper bound	Standardized reporting of split, resampling position, and preprocessing scope	Well-established
LLM-generated fraud	Adjacent benchmarks show 60–78 pt recall loss under rewriting; Fraud-R1 shows job inducements are hard	Deployed detectors face a distribution shift they were never tested for	Adversarial evaluation on job-fraud data; robust training methods	Emerging
Multilingual detection	1 direct study; EN–ZH gap in Fraud-R1	Regional fraud untouched by English-only models	Cross-lingual benchmarks; multilingual encoders	Emerging
Explainability beyond feature importance	Minority of studies provide any mechanism	Reviewer trust and regulatory compliance unaddressed	Counterfactual explanations for reviewers	Emerging
Federated or platform-shared detection	No direct evidence	Privacy-preserving cross-platform learning untested	Speculative; requires industry data agreements	Speculative

VIII. Future Research Agenda

The proposed studies are listed in Table 9 and the framework of the agenda is shown in Fig. 7. Every study can be traced back to the gap that it addresses via four elements: evidence of the gap, its consequences, the methodological need, and the proposed research. Priorities are based on evidence. A first multi-platform, temporally split, multilingual benchmark with

verified fraud ranks generated by LLMs is new, as many answers to other open questions are credible only on that basis. Standardized leakage-aware reporting (split protocol, resampling position, preprocessing scope, class distributions and minority-class metrics) is second, as it is relatively inexpensive and it alters the meaning of the numbers that are published. Third are validation studies carried out with the platform partners. They are difficult to collate and require the cooperation of industry which current publication incentives do not encourage.

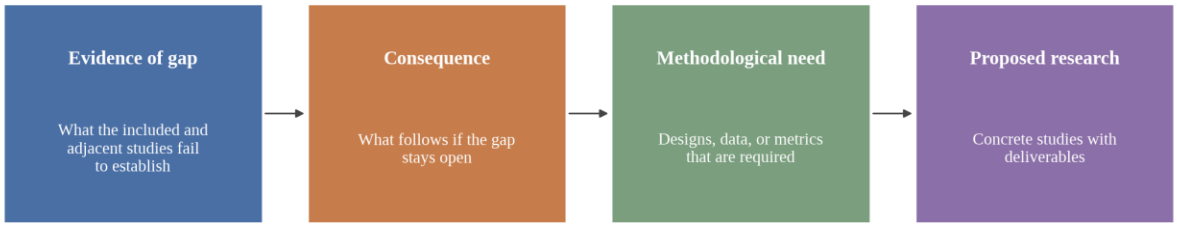


FIGURE 7. Evidence-based research agenda framework: each gap is traced from evidence to proposed study.

TABLE 9. Future research agenda (proposed studies).

Priority	Proposed study	Addresses	Deliverable
1	Construct and release a modern fake-job benchmark: 2023–2026 postings, multiple platforms, temporal splits, and a verified LLM-generated subset with provenance	Benchmark stagnation; LLM-era threat	Public dataset, evaluation protocol, and baseline suite
2	Standardized leakage-aware reporting checklist adopted across new studies (split, resampling position, preprocessing scope, class metrics)	Evaluation validity	Reporting checklist and template
3	Systematic within-study comparison of imbalance-handling methods on identical splits	Class imbalance practice	Benchmark comparison paper
4	Adversarial robustness evaluation of existing EMSCAD-trained detectors against LLM-rewritten postings	LLM-era threat	First direct robustness measurement
5	Multilingual and cross-lingual fraud detection benchmark (building on [30] and [26])	Multilingual gap	Cross-lingual evaluation suite
6	Platform-partnered deployment study with reviewer-in-the-loop metrics	Production validation	First published production validation

IX. Practical and Deployment Implications

This section is a conceptual one and not an empirical standard. The production data from any of the studies included in this review is not reported, so there is no production performance figure provided anywhere in this review. The following recommendation statements are the authors' and are not findings of the studies included.

First of all, decisions should be cost driven. One should use the calibrated probabilities [56] to set the threshold on the probability output of a detector, based on the cost of false positives, capacity of the reviewers and risk tolerance of the platform. There is no defensible universal threshold and calibration is not reported in the reviewed studies. Second, the dealing with the class imbalance shall be required automatically, and its impact shall be reported. Thirdly, there is no included study that shows temporal robustness, so a deployed system should presume drift [58] and measure it, using the most recent data as a rolling evaluation set. Fourth, explanations should be stronger if an action is taken based on a prediction than if the prediction is used for a posting that is placed in a queue for human review. Fifth, text generated by LLM should be presumed to be in traffic in real time. The adjacent domain evidence suggests that a detector that is only tested on clean human authored benchmarks will likely have a significantly poorer operating point on such text, but how much poorer is not known.

X. Limitations

There are five restrictions to the interpretation of the results. (1) No inter-rater agreement statistic could be reported as screening and risk of bias assessment was performed by one reviewer only without the presence of second reviewers. (2) The number of search results for each database was not stored, making it less reproducible; Appendix B provides a re-screening protocol. (3) This English language restriction does not apply to relevant non-English work in a multilingual problem domain. (4) Two studies were added to the main search through an addendum search, thus obscuring the sharp temporal limit of the search. (5) Synthesis of studies with different conditions of evaluation. However, there was limited quantitative comparison of results between studies, with the figures across studies being identified as being non-comparable, and extraction limited by the quality of reporting of the primary studies, which varies widely across studies.

XI. Conclusion

A review of 30 studies from 2017 to 2026 was analyzed. The field has taken the accuracy on one of the 10-year-old benchmarks to within a decade, and it's left the questions that count for deployment nearly unmeasured. One of the studies included external validation and none included temporal or production validation. The threat model for the benchmark has also shifted because generating models of convincing fraud text is cheap. There is some tentative evidence for a few conclusions. Both classical and transformer models perform

well when detecting fraud under benchmark conditions, while transformer encoders perform best in class-balanced metrics in the better-conducted studies. Ensemble and boosting methods are competitive at significantly lower costs. The impact of imbalance handling is measurable on minority class's performance. No evidence exists that any reported system could function on a current platform, operate in another language or in competition with an opponent using the same generative tools. The four steps are as follows: construct a modern benchmark, report evaluations that prevent leakage, measure generalization before claiming it and consider robustness to LLM-generated text as a property to be measured.

Appendix A. Data Extraction Fields

Information was extracted for each study that was included: bibliographic (authors, year, venue); dataset (name, size, fraud share, language, collection period, source platform, availability); features and modalities (text, metadata, structural, temporal, behavioral); model family and configuration; preprocessing; class-imbalance handling (method, location with respect to split); validation protocol; reported metrics (accuracy, precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix); baseline comparators; code or data availability; hyperparameter reporting; and stated limitations. If a study could not report a value, it was indicated as NR and not estimated. A full extraction sheet is available as a supplement.

Appendix B. Search Strategy and Reproducibility Protocol

Search will take place from January to March 2026. Databases: IEEE Xplore, ACM Digital Library, Google Scholar, ScienceDirect and arXiv. Block A (fraud target): “online recruitment fraud” OR “employment scam” OR “fake job posting” OR “fraudulent job advertisement” OR “job posting fraud.” Block B (method): "machine learning" OR "deep learning" OR "neural network" OR "natural language processing" OR "classification" OR "detection" OR "BERT" OR "transformer" OR "LSTM" OR "large language model". The two blocks were merged and the entire Boolean strings were modified to fit the field codes and syntax of the two databases.

The original search log did not contain per-database record counts, so Table 1 contains only stage totals. A re-screening process for replication consists of: (1) a secondary screening of each database-specific string with a date filter between January 2017 and March 2026 and record the number of hits per database in the screening log; (2) export the results and deduplicate them using a documented and defined tool and threshold; (3) perform a secondary screening of title/abstract and full-text and have two reviewers independently screen each string; (4) compute the Cohen's kappa for each string between the two independent reviewers; (5) extend the addendum search to the date the string was submitted.

Appendix C. Risk-of-Bias Rubric Decision Rules

Dataset adequacy: low = size, provenance, and class distribution all documented; unclear = partly documented; high = source unclear or size not stated. *Class-imbalance handling:* low = handling reported with its position relative to the split; unclear = handling mentioned without detail; high = not addressed. *Validation:* low = cross-validation [59], temporal split, or external dataset; unclear = single random split described; high = split not described. *Metric completeness:* low = accuracy plus precision, recall, and F1 or a precision-recall analysis; unclear = accuracy plus one class-balanced metric; high = accuracy alone or no extractable metrics. *Leakage risk:* low = the protocol description excludes all five mechanisms in Section V-D; unclear = partial description; high = protocol absent. *Reproducibility:* low = code or data released and hyperparameters reported; unclear = hyperparameters reported; high = neither. *Baseline adequacy:* low = comparison with at least one simple baseline; unclear = comparison only with other proposed models; high = none. The overall rating is the highest-risk rating among the seven dimensions (worst-case rule).

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RESEARCH INTERESTS

I am looking for research opportunities in cybersecurity, mainly where machine learning is used to detect attacks. I want to understand why intrusion detection models miss some attacks and raise false alarms on others, especially when the traffic data is noisy and heavily imbalanced. My intrusion detection project on CIC-IDS2017 is where this interest started.

EDUCATION

Bachelor of Science in Computer Science | Namal University Mianwali, Pakistan Oct 2022 – Jul 2026
CGPA: 3.19 / 4.00
Relevant coursework: Machine Learning, Artificial Intelligence, Information & Cybersecurity, Computer Networks, Cloud Computing, Data Structures & Algorithms, Compiler Construction, Databases

ACHIEVEMENTS

National Skill Competency Test (NSCT) | Higher Education Commission, Pakistan Apr 2026

- **Scored in the 98.5th percentile (top 1.5%) among 33,038 candidates** who appeared from universities across Pakistan in the HEC's first nationwide computing skills test.
- Test areas included programming, software engineering, databases, AI and machine learning, cybersecurity, and problem solving.

RESEARCH PROJECT

ML-Based Network Intrusion Detection System | CIC-IDS2017 dataset

- Developed an ML-based intrusion detection system for binary and multiclass attack classification on the CIC-IDS2017 network-traffic dataset.
- Compared Logistic Regression, Random Forest and Gradient Boosting in controlled experiments, evaluated with precision, recall, F1-score and ROC-AUC.
- Investigated class imbalance, feature importance, and false-positive and false-negative patterns to assess the practical challenges of ML-based intrusion detection.

PROJECTS

AI-Powered Disaster Management System | Final Year Project

- Built a full-stack disaster relief platform with Angular, Node.js and MongoDB, coordinating victims, NGOs and volunteers through 4 role-based dashboards.
- Developed an ML forecasting model that predicts resource demand 3 days ahead, and real-time disaster mapping with Leaflet.js.

Soybean Yield Prediction Model

- Built a regression model to predict soybean yield from agricultural traits, using feature engineering, exploratory data analysis, statistical analysis and visualizations to tune the model.

WORK EXPERIENCE

Software Engineering Intern | Speridian Technologies, Multan, Pakistan Jul 2025 – Aug 2025

- Built backend services and RESTful APIs with the .NET framework and C#.
- Built a frontend with Angular and TypeScript.
- Designed and optimized database schemas and queries in SQL Server.

COURSES & CERTIFICATIONS

- **Introduction to Cybersecurity** – Cisco (credly.com)
- **Introduction to IoT** – Cisco (credly.com)
- **Intro to Deep Learning** – Kaggle (kaggle.com)

TECHNICAL SKILLS

Programming: Python, C#, JavaScript, TypeScript, SQL
Machine learning & security: Machine learning, deep learning, intrusion detection, model evaluation (precision, recall, F1, ROC-AUC), digital image processing, cybersecurity and IoT fundamentals
Web, data & tools: Angular, RxJS, .NET, Node.js, Express.js, REST APIs, SQL Server, MySQL, MongoDB, Leaflet.js, Bootstrap, Git, Postman
Languages: English (professional working proficiency), Urdu (native)